Mohammed Aharrar Soulali

AI Engineer | Machine Learning
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Profile

Recent Master's-level engineering graduate from **École des Mines de Saint-Étienne**, specialized in **Data Science** and **Computer Science**. Projects in NLP, computer vision, and AWS, with a foundation in software engineering. **Available immediately**.

Experience

Final-Year Internship – Machine Learning Applied to Technical Reports — May 2025 – September 2025
EDF — *Paris, France*

- Designed an **NLP POC** on 17,000 French technical reports to predict importance levels and assist report writing
- Evaluated **TF-IDF**, **Word2Vec**, and **transformer** approaches for classification and similar-report suggestion

Assistant Engineer Internship — June 2024 – September 2024
École des Mines de Saint-Etienne Research Center — *Saint-Etienne, France*

- Developed a Python **computer vision** system for quality control on an automated pot-filling line
- Implemented Canny algorithms, ellipse detection, feature extraction, and optimization to estimate pot geometry

Education

Engineering Degree – Civil Engineer of Mines — September 2022 – September 2025
École des Mines de Saint-Étienne — *Saint-Etienne, France*

- **Data Science**: statistical learning | machine learning | time series | optimization
- **Computer Science**: software engineering | security and privacy | cloud & edge infrastructures
- **AI / Big Data**: knowledge representation | simulation | information systems | data visualization
- Minors: Image and Pattern Recognition | High-Performance Programming | Project Management

Bachelor's Degree in Mathematics — September 2023 – June 2024
Jean Monnet University — *Saint-Etienne, France*

- Mathematics degree completed in parallel with the engineering degree, strengthening foundations in analysis, algebra, probability and statistics.

Projects

Industrial Anomaly Detection | *Python, R, FPCA* — February 2025 – April 2025

- Analyzed production sound signals to detect anomalies in an automated filling process
- Benchmarked **PCA**, **FPCA**, **OSPCA**, and Mahalanobis methods; best F1-score: **0.92**

Industrial Video Analysis with Orano | *Python, Computer Vision* — September 2024 – January 2025

- Automated inspection-video analysis to quantify powder build-up in pneumatic transport tubes
- Used image filtering and geometric indicators to locate affected areas and provide a quality diagnosis

IoT Cloud Pipeline | *AWS, Java, S3, SQS, Lambda* — November 2024 – January 2025

- Designed an AWS pipeline to ingest, process, and consolidate IoT data
- Compared server-based and serverless architectures using **S3, SQS, EC2, Lambda**, and CloudWatch

Multi-Agent Traffic Optimization | *Java, Simulation, AI* — January 2024 – June 2024

- Built a Java simulation of an intersection with intelligent traffic lights to reduce vehicle waiting time
- Implemented decentralized decisions based on cumulative waiting time; reduced delays by a factor of **2 to 4** depending on scenarios

Languages

French: B2 **English**: C1 (TOEIC 945, 2024) **Arabic**: C1 (Native language) **German**: A1+

Skills

Programming: Python, Java, SQL, PostgreSQL, R, Bash/Shell
ML / AI: scikit-learn, TensorFlow, NLP, Transformers, RAG, LangChain, FAISS, Ollama, LSTM, Computer Vision
Cloud / DevOps / Backend: AWS, Flask, FastAPI, REST APIs, Git/GitHub, Docker, CI/CD
Other: Scrum, Problem-solving, Teamwork, Adaptability